

Anuj Chauhan

Ph.D. National Taiwan University (國立臺灣大學) · Taipei City, Taiwan

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Taiwan Employment Gold Card Holder (台灣就業金卡)

PROFESSIONAL SUMMARY

Industrial Simulation Engineer delivering production-validated **physics-based simulation** solutions at Pegatron Corp. I design and deploy **CFD (fluid dynamics)** and **FEA (stress, deformation)** models across the full ANSYS suite — **Fluent, Mechanical, Icepak, LS-DYNA**, and **Twin Builder** — to solve real thermal, structural, multiphase, and **data center cooling** challenges in high-reliability hardware. Specialized in building **Digital Twins** and **Reduced Order Models (ROMs)** that turn complex 3D simulations into rapid, decision-ready insights. Backed by a **Ph.D. from National Taiwan University**, hands-on semiconductor nanofabrication expertise (EBL, photolithography, RIE/ICP, full cleanroom integration). Bridging the rare gap between **nanoscale fabrication** and **industrial-scale simulation** — ready to apply both to advanced semiconductor manufacturing.

CORE COMPETENCIES & TECH STACK

Semiconductor Fabrication & Metrology

- **Lithography:** Electron Beam Lithography (EBL), Advanced Photolithography, PR Optimization, CD Control.
- **Cleanroom Processing:** End-to-end Process Integration, PVD/CVD, ALD, Plasma Etching (RIE/ICP), Wet Processing.
- **Metrology:** SEM, TEM, Optical Inspection, Defect Analysis, Yield Optimization, Critical Dimension (CD) Validation.

☞ Process Simulation & Modeling

- **Focus:** Multi-Physics Modeling (Thermal/Stress), Data Center Cooling & Thermal Management, Structural Reliability, Physics-Based Digital Twins, DOE, ROM.
- **Stack:** ANSYS (Mechanical, Fluent Multiphase VOF, Icepak (Beginner), Heatsink, LS-DYNA, Twin Builder, EnSight: PostProcessing), PyTwin, NVIDIA Omniverse, PyAnsys, CAD.
- **Industry Proven:** All simulation capabilities applied and validated on real industrial projects at Pegatron Corp — from concept to production-ready digital twin deployment.

PROFESSIONAL EXPERIENCE

PEGATRON CORPORATION

Section Manager, Software R&D

Technical Manager, Software R&D

Taipei, Taiwan

Mar 2025 – Present

Jun 2024 – Mar 2025

- **Multi-Physics Simulation & Reliability:** Executed advanced ANSYS Mechanical and Fluent simulations for industrial projects, performing high-fidelity stress, deformation, and CFD analysis to optimize structural integrity and fluid flow.
- **LS-DYNA Simulation:** Developed non-linear FEA models in LS-DYNA for large-scale deformation and crash analysis, integrating simulation outputs directly into digital twin workflows.
- **Data Center Thermal & Cooling Simulation:** Developed high-fidelity CFD models using **ANSYS Fluent** and **Icepak** to simulate airflow, heat dissipation, and cooling efficiency across data center rack and server environments; delivered actionable insights for thermal management optimization and energy efficiency improvements.
- **Reduced Order Modeling (ROM):** Engineered computationally efficient models using **ANSYS Twin Builder** to compress complex 3D FEA/CFD results for rapid, accurate reliability assessment.
- **Digital Twin Pipeline:** Spearheaded development of physics-based simulation workflows enabling predictive monitoring of industrial hardware performance under varying structural and thermal constraints.

NATIONAL TAIWAN UNIVERSITY (Grad. Institute of Photonics & Optoelectronics)

Postdoctoral Fellow

Taipei, Taiwan

Apr 2023 – May 2024

- **Advanced Lithography & Nanofabrication:** Designed and patterned ultra-compact plasmonic waveguides and optical ring-resonators, achieving strict nanoscale tolerances using **Electron Beam Lithography (EBL)** and precision **Photolithography**.
- **Process Integration & Optimization:** Directed comprehensive cleanroom process flows (Deposition, Patterning,

Etching). Optimized photoresist profiles and plasma etching recipes (RIE/ICP) to significantly improve device yield and reliability.

- **Metrology & Characterization:** Conducted rigorous inspection using **SEM** and optical metrology tools to validate Critical Dimensions (CD) and ensure high process fidelity across integration cycles.

SHARDA UNIVERSITY (*Department of Physics & Environmental Sciences*)
Research Assistant

Greater Noida, India
Jan 2018 – Dec 2018

- Fabrication and characterization of **dye-sensitized solar cells** and **supercapacitors**; developed hands-on materials processing and electrochemical characterization skills.

EDUCATION

National Taiwan University

Ph.D., Chemistry

Award: NTU Outstanding International Graduate Student Scholarship

Taipei, Taiwan

2019 – 2023

Gautam Buddha University

M.Sc., Applied Physics

Greater Noida, India

2014 – 2016

University of Delhi

B.Sc., Physics

New Delhi, India

2010 – 2013

SELECTED PUBLICATIONS

1. Yu-Chuan Chu, ... **Anuj Chauhan**, et al., "Performance Comparisons of GaN Vertical Transistors with Sidewalls Treated by TMAH and H₃PO₄ Solutions," *IEEE Electron Device Letters*, Vol. 45, 2024.
2. Rohit Gupta, ... **Anuj Chauhan**, et al., "Surface Acoustic Wave Actuated Plasmonic Signal Amplification in a Plasmonic Waveguide," *Discover Nano*, Vol. 19, 2024.
3. **Anuj Chauhan**, et al., "Photocatalytic Degradation-based Efficient Elimination of Pesticides using Ruthenium/Gold Metal Nanoparticle-anchored Zirconium Dioxide," *New Journal of Chemistry*, RSC Vol. 46, 2022.
4. J. H. Lai, ... **Anuj Chauhan**, et al., "Antibiotic Drugs Removal by Visible Light-driven Photocatalysis Using Pt/Ru Nanoparticle-Decorated Hafnium Oxide Nanohybrids," *Journal of Env. Chem. Eng.*, Elsevier, 2022.
5. N. Dhenadhayalan, **Anuj Chauhan**, et al., "Architecting 3D Prism Shaped Ge/GeO Nanomaterial for Photocatalytic Degradation of Pendimethalin and Dinotefuran Pesticides," *Materials Today Chemistry*, Elsevier, 2022.
6. **Anuj Chauhan**, et al., "Evanescent Wave Cavity Ringdown Spectroscopy Based Interfacial Sensing of Prostate-Specific Antigen," *Sensors and Actuators B*, Elsevier, 2021.

REFEREES

Prof. Jian-Jang Huang (Ph.D. Advisor)

Professor, National Taiwan University

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Prof. Toshio Kasai

Professor, National Taiwan University

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